Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. We trained each model for 20 epochs with a batch size of 32.

2 Results

Figure 1 shows the validation accuracy by dataset.

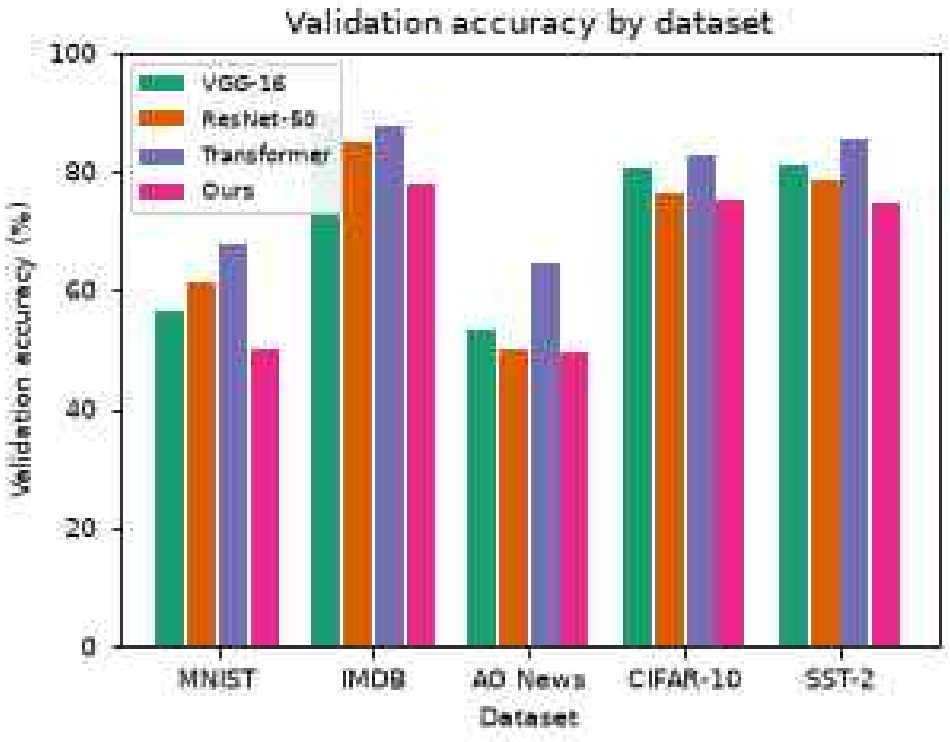


Figure 1: Validation accuracy by dataset for the models in this experiment.

On IMDB, Ours obtains a validation accuracy of over 75% (Figure 1). Ours reaches 54.6% validation accuracy on SST-2 (Figure 1). On MNIST, ResNet-50 obtains a validation accuracy of 71.2%. Transformer achieves 68.0% validation accuracy on MNIST.

Figure 2 shows the validation accuracy of LSTM by dataset.

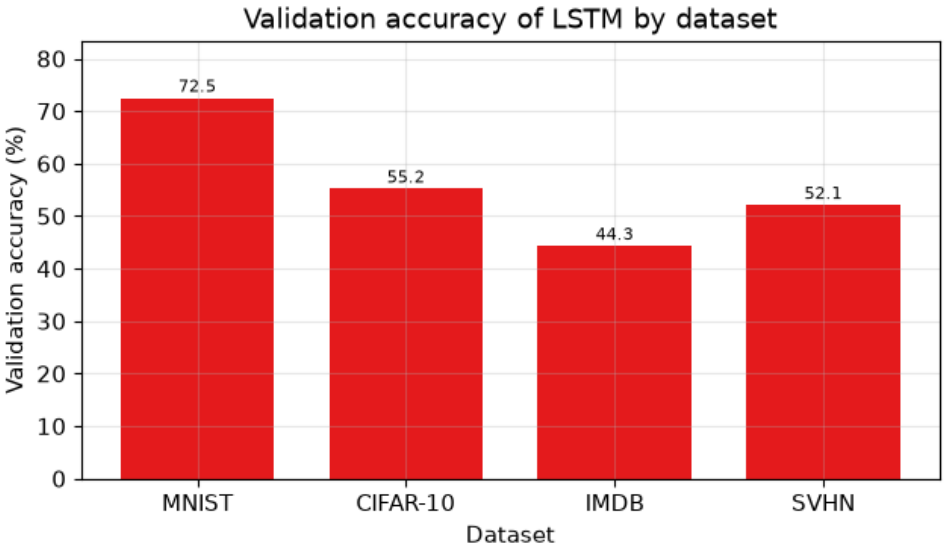


Figure 2: Validation accuracy of LSTM by dataset for the models in this experiment.

On IMDB, LSTM obtains a validation accuracy of 44.3%. On IMDB, LSTM obtains a validation accuracy of 2.6%. LSTM scores 17.3 points higher on MNIST than on CIFAR-10.

Figure 3 shows the accuracy by dataset.

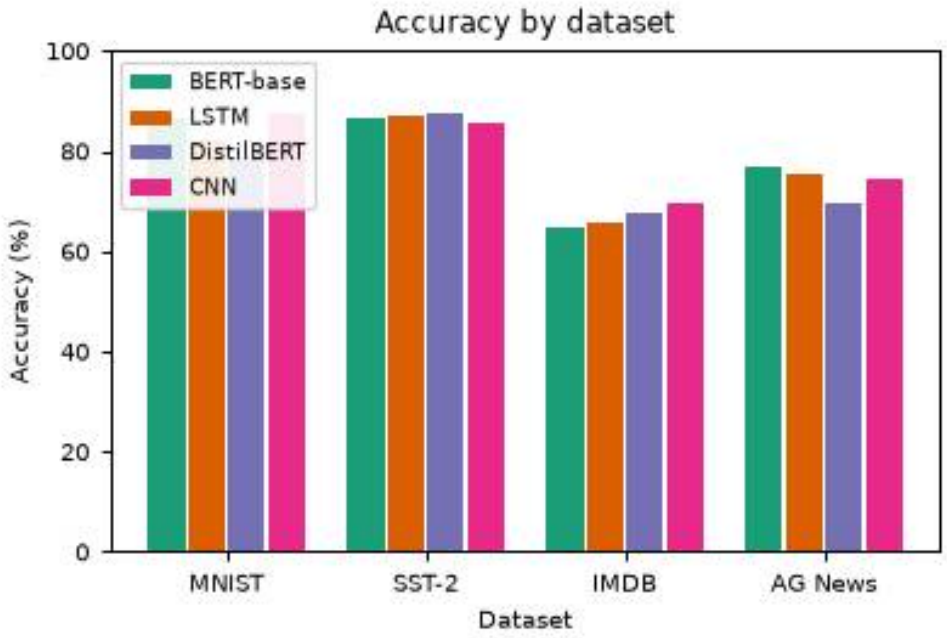


Figure 3: Accuracy by dataset for the models in this experiment.

On MNIST, BERT-base is 4.3 points better than LSTM (Figure 3). As shown in Figure 3, BERT-base reaches 77.0% accuracy on AG News. BERT-base achieves 75.2% accuracy on AG News (Figure 3). On IMDB, LSTM obtains a accuracy of 55.9%.

Figure 4 shows the test accuracy by dataset.

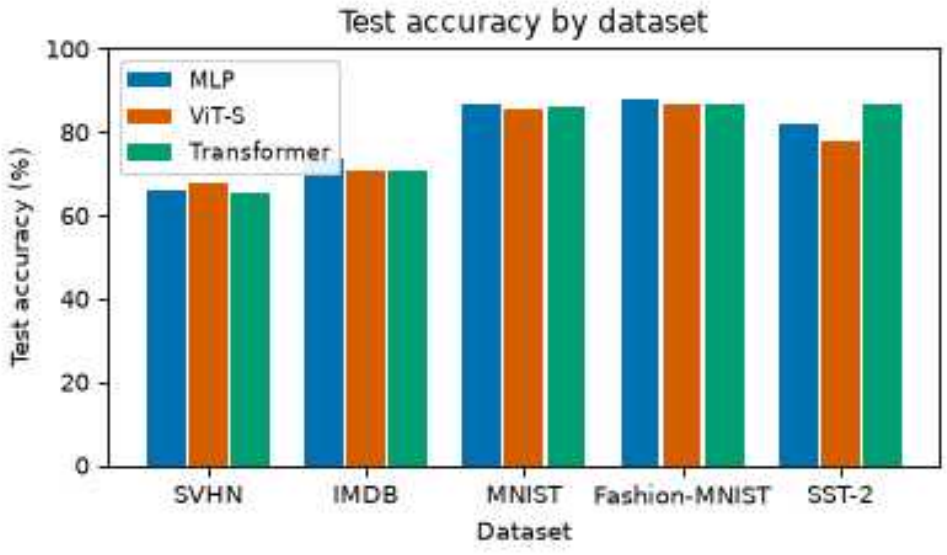


Figure 4: Test accuracy by dataset for the models in this experiment.

ViT-S outperforms Transformer by 52.4 percentage points on SVHN (Figure 4). MLP achieves 71.0% test accuracy on SVHN. MLP reaches 87% test accuracy on MNIST (Figure 4). As shown in Figure 4, on SVHN, MLP obtains a test accuracy of 67.6%.

3 Discussion

We note that the training was stable for all models.